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Human-Algorithm Collaboration with Private Information: Naïve Advice-Weighting Behavior and Mitigation

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Abstract. Even if algorithms make better predictions than humans on average, humans may sometimes have private information that an algorithm does not have access to that can improve performance. How can we help humans effectively use and adjust recommendations made by algorithms in such situations? When deciding whether and how to override an algorithm's recommendations, we hypothesize that people are biased toward following naïve advice-weighting (NAW) behavior; they take a weighted average between their own prediction and the algorithm's prediction, with a constant weight across prediction instances regardless of whether they have valuable private information. This leads to humans overadhering to the algorithm's predictions when their private information is valuable and underadhering when it is not. In an online experiment where participants were tasked with making demand predictions for 20 products while having access to an algorithm's predictions, we confirm this bias toward NAW and find that it leads to a 20%–61% increase in prediction error. In a second experiment, we find that feature transparency—even when the underlying algorithm is a black box—helps users more effectively discriminate how to deviate from algorithms, resulting in a 25% reduction in prediction error. We make further improvements in a third experiment via an intervention designed to move users away from advice weighting and instead, use only their private information to inform deviations, leading to a 34% reduction in prediction error.

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Keywords: human-algorithm interaction • forecasting • behavioral operations • algorithm transparency • advice taking

1. Introduction

Organizations are seeking to incorporate data-driven algorithms into their decision-making processes. PricewaterhouseCoopers (2022) reports that 86% of executives consider artificial intelligence (AI) algorithms a "mainstream technology," with 74% believing that these algorithms would add value to their companies by, for example, improving operations, marketing, and human resources (HR) decisions. However, despite the promise of algorithms and their widespread adoption, implementations are not always successful. In fact, only 10% of companies that adopted advanced algorithms, like AI, report seeing significant financial gains according to Ransbotham et al. (2020). When algorithms show significant promise in simulations and on historical data but have disappointing performance once implemented, blame is commonly aimed at the humans that use the algorithms. This blame is sometimes warranted; people often have the power to override algorithm recommendations, and there are many examples of such overrides degrading performance (see Section 2.1).

Given such disappointing examples, it can be tempting to take the perspective that humans are a *barrier* to achieving the benefits of algorithms. If algorithm aversion causes people to override superior algorithms, should the goal not be to get people to trust algorithms more or even remove them from the process? Indeed, finding ways to increase people's adherence to and trust in algorithms has been the focus of much research (see Section 2.3). Although this response is appropriate in some cases, we take a different and more collaborative perspective. Namely, we start with the premise that humans are not merely barriers but that humans and algorithms have relative strengths and weaknesses. Even when algorithms are superior to humans on average, theoretically, their collaboration should be

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able to outperform either on its own. In settings where this is possible, what is it about human overriding behavior that causes them to miss this opportunity? How can we design the interaction between humans and algorithms such that their collaboration is more successful?

To investigate these questions, we focus on prediction tasks (e.g., forecasting demand) where human decision makers are given recommendations in the form of predictions from an algorithm, which they are freely able to override to make a final prediction. We consider key relative strengths of humans and algorithms. Prediction algorithms advance in accuracy and sophistication every year, so we simplify their strength by assuming that an algorithm uses its available information optimally. Relative to algorithms, humans are noisy and more limited in information processing power. So, what advantages do humans have? One of their most important relative strengths is that humans sometimes have access to *private information*: any information with predictive value that the algorithm does not take into account. There are many examples. A fashion retail manager may know from social media that a product is trending, but such information may not be used by the company's forecasting algorithm (Cui et al. 2018). A doctor may know that a patient's surgery is unusually complicated based on how it looks, even if such information is not in the hospital's information system (Ibrahim et al. 2021, Kim and Song 2022). HR managers making hiring decisions interview candidates, although a scoring algorithm may be based only on test scores and resumes (Hoffman et al. 2018). In such examples, companies often cite private information as one of the key reasons why final decision authority is given to employees (Davenport and Miller 2022). Of course, having valuable private information does not necessarily mean that humans will make override decisions that improve upon the algorithm's performance. But, eliminating humans from the process precludes being able to take advantage of such private information.

Focusing our attention on these relative strengths allows us to approach our research questions in the context of an information aggregation problem, which allows us to leverage existing decision analysis, judgment, and decision-making literature to better theorize how and why overriding behaviors tend to be biased. To develop our theory, we construct a mathematical model that defines the information setting, the algorithm's predictions, and the best prediction benchmark. We then examine the impact of certain behavioral assumptions about how people make final predictions after seeing an algorithm's recommendation. We theorize that because of cognitive limitations, people are biased toward following a predictable heuristic that we call *naïve advice weighting* (NAW). A person who follows NAW arrives at a final prediction by taking a constant weighted average between what the algorithm recommends and what their own prediction would have been without the algorithm. Although such an advice-weighting strategy is natural and commonly recommended as an effective way to aggregate advice from multiple sources, we show that NAW is typically suboptimal when providing an algorithm's predictions to a human with private information. Namely, NAW is overly constant; it causes people to overadhere to the algorithm when they have highly valuable private information and underadhere to the algorithm when they do not. Moreover, it is suboptimal because the best prediction may not even lie between the algorithm's recommendation and the person's own initial prediction; the person's initial prediction may be lower than the algorithm even when their private information suggests predicting higher than the algorithm.

We further conduct a controlled online experiment that seeks to test the over- and underadherence pattern predicted by our theory. In Study 1, following training and feedback about the algorithm's and their own performances, participants predict demand for 20 products with the algorithm's recommendations. The only difference between conditions is that some participants always have high-impact (very valuable) private information, some always have low-impact private information, and some face a mixed set of the two instances. We find that participants who always have low-impact private information generally adhere to the algorithm, whereas those who always have high-impact private information generally do not. However, consistent with NAW, participants seeing a mixed set adhere to the algorithm to a similar degree across high- and low-impact private information instances, resulting in the predicted over- and underadherence pattern. This leads to a 20%–61% increase in prediction error relative to the nonmixed conditions. In summary, participants in the mixed set condition failed to use their private information to collaborate effectively with the algorithm because they could not differentiate when their private information was valuable.

Based on these results, we design and test a type of algorithm transparency aimed at mitigating the underlying driver of bias. Specifically, we hypothesize that *feature transparency*—explicit training on the variables that the algorithm takes into account—helps participants address the core problems of NAW by making it easier for them to identify what their private information is, when it warrants a substantial deviation from the algorithm, and in which direction. In Study 2, we compare *feature transparency* with *no transparency*. Using the same mixed set condition from Study 1, Study 2 shows that *feature transparency* indeed helps people detect when they should adhere more or less to the algorithm, resulting in a 25% reduction in prediction

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error. In Study 3, we design an intervention aimed at increasing the effectiveness of *feature transparency* by nudging people away from advice-weighting strategies altogether, thus making it more likely that they adjust the algorithm's predictions solely using their key relative strength: their private information. We find that this intervention is highly effective, improving the rate of adjustments in the right direction and leading to an additional 21% reduction in prediction error over *feature transparency* alone.

We summarize our main contributions as follows.

1. We define and examine a new theory that describes algorithm overriding behavior when people have private information. We propose and find empirical evidence that people are biased toward a *naïve advice-weighting* heuristic.

2. We analyze how NAW degrades human-algorithm collaborative performance, finding both theoretically and empirically an over- and underadherence pattern that depends on the value of people's private information. We illustrate that the cost of these biases can be significant.

3. We design and experimentally test *feature transparency* as an implementable mitigation approach that can help people better identify and use their private information. We achieve further improvements by nudging people away from advice-weighting heuristics and instead, toward using only their private information to adjust algorithmic recommendations.

2. Literature Review

2.1. Algorithm Overriding

Do human overrides to algorithm predictions help or hurt in practice? Field evidence from a variety of business contexts provides several examples of overriding degrading performance; doctor overrides of task-scheduling algorithms decreased productivity (Ibanez et al. 2018), retail store manager overrides of price markdown algorithms decreased revenue (Caro and Sáez de Tejada Cuenca 2023), and auto-part manager overrides of stock keeping unit phase-out algorithms decreased profits (Kesavan and Kushwaha 2020). Laboratory experiments provide further examples (e.g., see Lehmann et al. 2022 and Snyder et al. 2024). Of course, human overrides do not *always* degrade performance. For example, in Fildes et al. (2009), forecaster overrides improved accuracy on average in three of four supply chain companies investigated. Moreover, even in settings where human overrides degrade performance on average, they may improve performance on predictable subsets of situations. For example, in Kesavan and Kushwaha (2020), although overrides hurt profits on average, they improve profits for growth-stage products. Similarly, for demand forecasters in Khosrowabadi et al. (2022), algorithm overrides did not improve accuracy on average but did help for expensive and nonfresh products. In general, these field studies point to the importance of understanding *how* people make override decisions so that we can help humans override in such a way that consistently yields improvement over the algorithm alone.

Relatedly, there is diverse research that suggests a variety of possible *psychological* factors contributing to why people override algorithms. For example, even if an algorithm performs well, people may be averse to using something that they do not understand (Yeomans et al. 2019). They might feel like the task is too subjective for an algorithm (Castelo et al. 2019). People may be more tolerant of their own mistakes than an algorithm's mistakes (Dietvorst et al. 2015). They may also perceive an algorithm as too simple or too complex (Lehmann et al. 2022). In contrast to this body of work on psychological reasons for overriding, we focus on a setting with a purely *rational* reason for overriding; people may have access to private information that the algorithm does not use.

2.2. Aggregation Strategies

How *should* one aggregate an algorithm's predictions with a human who has private information? Such a prescriptive question belongs to a broader area of research on judgment aggregation strategies, which has addressed this question primarily in the context of aggregating multiple human judgments. A main finding is that simple averaging strategies work surprisingly well (e.g., Clemen 1989, Blattberg and Hoch 1990, Surowiecki 2005, Larrick and Soll 2006). However, substantial improvements over simple averaging can be achieved when people have shared information. Recent research has developed strategies for combining judges' predictions to address this shared information problem: for example, by asking an additional question that can help an algorithm infer the amount of shared information (e.g., Palley and Soll 2019). Other strategies include strategic identification of experts and upweighting their predictions (e.g., Soule et al. 2024). Our work similarly seeks to understand and address the shared information problem. However, unlike most of the above papers, it addresses the problem between *an algorithm* and a human, importantly considering the relative strengths and weaknesses of each.

In a review paper, Arvan et al. (2019) outline two paradigms of how human and algorithm forecasts can be combined to produce superior forecasts. In one paradigm, an algorithm makes the final forecast after receiving a human's prediction as an input. Several papers study how best to elicit forecasts and/or private information from humans and how to use them as inputs to algorithms (e.g., Ball and Ghysels 2018, Flicker 2018, Ibrahim and Kim 2019, Ibrahim et al. 2021, Brau et al. 2023). An alternate paradigm, which

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we study, is when a human makes the final forecast (e.g., see Luong et al. 2020, although their focus is on binary—rather than continuous—prediction tasks). Such a human-oversight policy is common in practice as it allows humans to override the algorithm's recommendation when they have private information (Davenport and Miller 2022). It is even required across many settings for legal or ethical reasons (Green 2022).

This second paradigm is related to a broader stream of literature on how people make final decisions when they receive a recommended prediction from an external advisor. A common model used to study this question is a judge-advisor system (JAS), which can be applied to human advisors (e.g., Bonaccio and Dalal 2006, Soll et al. 2022) and algorithmic advisors (Logg et al. 2019, Lehmann et al. 2022). In JAS, a human judge first forms their initial prediction; then, they receive a prediction from an advisor, and they make a final prediction. The weight on advice (WOA) metric refers to where the final prediction lies on the interval between the initial prediction and the advice; a WOA of zero represents ignoring the advice, whereas a WOA of one represents completely taking the advice. Many papers report on various factors that impact WOA, but all typically report very high rates (often over 95%) of what we call "advice-weighting" behavior, where people's final predictions are a weighted average of their initial prediction and the advice (e.g., Gino and Moore 2007, Soll and Larrick 2009, Logg et al. 2019). This is for good reason; in cases without clear private information, it can be difficult to identify a strategy that will outperform advice weighting. We examine how people take the advice of an algorithm and also, report WOA measurements. However, unlike most papers in this stream of work, we examine settings in which the participant has objective and measurable reasons to deviate from the algorithm's advice—namely, valuable private information—that may vary across instances. Moreover, we examine how participants can improve their performance by deviating from an advice-weighting approach using their private information.

Beyond the aggregation strategies detailed above, delegation strategies have also been proposed for settings where humans sometimes have valuable private information (Ross and Taylor 2021); such strategies typically implement a heuristic that delegates prediction instances to either a fully automated algorithm or to the employee (Fügener et al. 2022). Naturally, in order for delegation strategies to work well, the system designer must be able to predict when the employee has valuable private information *often without observing it*, which—as in our experiments—is not always possible. Furthermore, final decision-making authority should or must stay with a human in some situations for a variety of other reasons (e.g., because of regulatory, ethical, or oversight considerations).

2.3. Algorithm Transparency

How do you design algorithms to mitigate end-user biases? Although there are a variety of strategies, our paper focuses on providing ex ante algorithm transparency, where aspects of the model are described to the user. There are many such types of transparency. Comprehensive ex ante transparency allows people to fully *simulate* the algorithm's predictions (Lage et al. 2019). One can also provide transparency into certain components, such as how the model was trained (Anik and Bunt 2021, Gebru et al. 2021); these types of component transparency have been advocated and implemented by industry leaders, like Google and IBM (Mitchell et al. 2019, Hind et al. 2020, Gebru et al. 2021). However, unlike the above papers, we make precise predictions about how private information interacts with human cognitive biases to make different types of transparency help or hurt in predictable circumstances.

Commonly, algorithm transparency is provided along with a proalgorithm nudge in an attempt to increase end-user trust, which can be subjectively measured (e.g., a Likert scale as in Cadario et al. 2021) or observed (e.g., an algorithm use rate as in Yin et al. 2019). Nevertheless, there is not uniform evidence that algorithm transparency and nudging people to follow the algorithm increase trust (Bolton et al. 2022, Lehmann et al. 2022). Also, increased trust does not always lead to better outcomes as people may suffer from information overload (Poursabzi-Sangdeh et al. 2021) or overly trust the algorithm when they should not (Lakkaraju and Bastani 2020). We contribute to these papers by identifying a type of transparency and a corresponding nudge that increase trust when the algorithm should be superior and decrease trust otherwise.

3. Theory Development

3.1. Model Setting, Definitions, and Assumptions

Consider a setting where outcome Y_i is a function of public feature vector $\boldsymbol{X}_i^{pub}$, private feature vector $\boldsymbol{X}_i^{priv}$, and mean-zero random noise ϵ_i for each instance $i = 1, 2, \ldots, n$: that is,

$$Y_i = f_{actual}(\boldsymbol{X}_i^{pub}, \boldsymbol{X}_i^{priv}) + \epsilon_i\,. \tag{1}$$

We assume $\{(\boldsymbol{X}_i^{pub}, \boldsymbol{X}_i^{priv}, \epsilon_i)\}_{i=1,..,n}$ are independent and identically distributed. (However, $\boldsymbol{X}_i^{pub}$ and $\boldsymbol{X}_i^{priv}$ can be dependent.)

The human decision maker is tasked with predicting outcome Y_i given a realization of the feature vector $(\boldsymbol{x}_i^{pub}, \boldsymbol{x}_i^{priv})$. We model human j's prediction without seeing an algorithm, which we call their *initial prediction*: for instance, i as

$$\hat{Y}_{ij}^{init} = f_{init,j}(\boldsymbol{X}_i^{pub}, \boldsymbol{X}_i^{priv}) + \eta_{ij}\,. \tag{2}$$

Here, $f_{init,j}$ may differ from f_{actual}, reflecting the idea that humans may use feature information in a systematically

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incorrect manner. Also, η_{ij} is a zero-mean random variable, reflecting the idea that even given identical information, humans may be noisy (e.g., Su 2008, Kahneman et al. 2021).

We consider the situation where the decision maker has access to an algorithm that they can use to help predict Y_i. The algorithm uses only public features as inputs. Its prediction is

$$\hat{Y}_i^{alg} = f_{alg}(\boldsymbol{X}_i^{pub}). \tag{3}$$

Finally, we can define the *best prediction benchmark* as

$$Y_i^* = f_{actual}(\boldsymbol{X}_i^{pub}, \boldsymbol{X}_i^{priv}). \tag{4}$$

The gap between the best prediction benchmark Y_i^* and the algorithm's prediction $\hat{Y}_i^{alg}$ represents the potential improvement that the human could theoretically make over the algorithm's prediction, which could come from either (i) the use of private features $\boldsymbol{X}_i^{priv}$, for which the algorithm does not have access, and/or (ii) better use of public features $\boldsymbol{X}_i^{pub}$. For ease of exposition and in line with our experiments, we assume that the algorithm optimally uses $\boldsymbol{X}_i^{pub}$ when making its predictions, and thus, any improvement that the human can make over the algorithm's prediction is because of their use of $\boldsymbol{X}_i^{priv}$. With this in mind, we define the *impact of the private features* as follows.

Definition 1. The impact of the private features for feature vector $(\boldsymbol{X}_i^{pub}, \boldsymbol{X}_i^{priv})$ is

$$V_i \triangleq Y_i^* - \hat{Y}_i^{alg} = f_{actual}(\boldsymbol{X}_i^{pub}, \boldsymbol{X}_i^{priv}) - f_{alg}(\boldsymbol{X}_i^{pub}).$$

Because realization v_i can be positive or negative, we will use the squared impact of the private features $(V_i)^2$ as a measure of the impact in an absolute sense. The following two assumptions are sufficient but not always necessary for our results.

Assumption 1. *The human's initial prediction and the algorithm's forecasts are unbiased (i.e.,* $\mathbb{E}[\hat{Y}_i^{init} - Y_i] = \mathbb{E}[\hat{Y}_i^{alg} - Y_i] = 0$*).*

Note that the expectation is over random variables $\boldsymbol{X}_i^{pub}, \boldsymbol{X}_i^{priv}, \eta_{ij}$, and ϵ_i. It does *not* imply that the human and algorithm are unbiased for every instance i; rather, it implies that the human and algorithm will not be systematically too high or low over a large number of instances.

Assumption 2. *The human's initial prediction error* $\hat{Y}_i^{init} - Y_i$ *and the algorithm's prediction error* $\hat{Y}_i^{alg} - Y_i$ *are independent.*

Assumption 2 states that the human's initial prediction accuracy does not provide information about the algorithm's accuracy and vice versa.

3.2. Behavioral Model of Advice Weighting

How do humans with access to an algorithm's prediction make a final prediction? Following the advice-weighting literature (see Section 2.2), we hypothesize that humans tend to take a weighted average of their initial prediction and the algorithm's prediction to make a final prediction:

$$\hat{Y}_{ij}^{final} = \lambda_{ij}\hat{Y}_i^{alg} + (1 - \lambda_{ij})\hat{Y}_{ij}^{init}, \tag{5}$$

where $\lambda_{ij} \in [0, 1]$ is the weight that human j places on the algorithm's prediction. For brevity, we omit subscript j when the context is clear. A larger (smaller) value of λ_i means that the human places more (less) weight on the algorithm's prediction.

One can see that the human's final prediction accuracy depends, in part, on how they choose the value of λ_i. We believe that humans tend to be biased toward *naïve advice weighting*, which we define as placing the same weight on the algorithm's prediction regardless of feature vector $(\boldsymbol{x}_i^{pub}, \boldsymbol{x}_i^{priv})$ (i.e., $\lambda_i = \lambda$ for each instance i). Naïve advice weighting is suboptimal for two reasons.

- First, differential weights are not applied across instances with different impacts of the private features. We examine this driver of suboptimality in Section 3.3. To do so, we contrast naïve advice weighting with *sophisticated advice weighting*—where differential weights are applied depending on feature vector $(\boldsymbol{x}_i^{pub}, \boldsymbol{x}_i^{priv})$.
- Second, advice weighting in general—even the best-possible sophisticated advice-weighting strategy—is suboptimal when Y_i^* cannot be characterized as a convex combination of $\hat{Y}_i^{init}$ and $\hat{Y}_i^{alg}$; we examine this driver of suboptimality in Section 3.4. Importantly, the human could use their private information to identify when advice weighting is suboptimal.

Our results will guide our experimental designs and hypotheses, which we preview in Section 3.5.

3.3. Naïve Advice Weighting Is Suboptimal Because the Weights Are Overly Constant

We now show that a sophisticated advice-weighting strategy that implements nonconstant weights can outperform even the best-possible naïve advice-weighting strategy found by solving

$$NAW: \quad \min_{\lambda \in [0,1]} \mathbb{E}[(Y_i - (\lambda\hat{Y}_i^{alg} + (1 - \lambda)\hat{Y}_i^{init}))^2]. \tag{6}$$

NAW finds the constant weight on the algorithm that minimizes the expected squared error of the final prediction, $\hat{Y}_i^{final}$. Practically, it could be approximated by solving for the λ that minimizes the sum of squared errors over a large historical data set of realizations $\hat{y}_i, \hat{y}_i^{alg}, \hat{y}_i^{init}$.

Proposition 1. *Under naïve advice weighting, the optimal weight on algorithm* λ^{NAW} *is*

$$\lambda^{NAW} = \frac{\mathbb{E}[(Y_i - \hat{Y}_i^{init})^2]}{\mathbb{E}[(Y_i - \hat{Y}_i^{init})^2] + \mathbb{E}[(Y_i - \hat{Y}_i^{alg})^2]}.$$

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The optimal weight on the algorithm under NAW is intuitively appealing; it puts more weight on the algorithm if it has smaller expected squared errors relative to the human's initial forecasts. Empirical evidence also suggests that humans generally weight advice in a similar manner; they increase their weight on advice as the advisor's accuracy increases (Harvey and Fischer 1997, Yaniv and Kleinberger 2000, Sniezek and Van Swol 2001, Sah et al. 2013, Soll et al. 2022).

Although intuitive, naïve advice weighting is suboptimal in part because the weight is constant, but the algorithm's accuracy is not when the impact of the private features varies. Intuitively, the human could use (x_i^{pub}, x_i^{priv}) to predict when the algorithm will be more (less) accurate and put more (less) weight on the algorithm. To formalize this intuition, we define a *sophisticated advice weighter* as one who first categorizes feature vectors into groups with different average impacts of the private features and then, calculates a different weight on algorithm for each group. (We consider only two groups, but the analyses could be extended to more.) Let $\mathcal{S}$ denote the domain of (X_i^{pub}, X_i^{priv}). A sophisticated advice weighter can partition $\mathcal{S}$ into sets $\mathcal{S}_L, \mathcal{S}_H$ such that

$$E[(V_i)^2 | (X_i^{pub}, X_i^{priv}) \in \mathcal{S}_L] < E[(V_i)^2 | (X_i^{pub}, X_i^{priv}) \in \mathcal{S}_H].$$

Thus, $\mathcal{S}_L$ has a lower expected squared impact of the private features, whereas $\mathcal{S}_H$ has a higher expected squared impact of the private features. Given such partitions, a sophisticated advice weighter (SAW) can solve the following problem to determine the optimal weights, λ_L^{SAW} and λ_H^{SAW}, in each partition:

$$SAW: \min_{\lambda_L, \lambda_H \in [0,1]} \sum_{k=L,H} \mathbb{E}[(Y_i - (\lambda_k \hat{Y}_i^{alg} + (1-\lambda_k)\hat{Y}_i^{init}))^2 | (X_i^{pub}, X_i^{priv}) \in \mathcal{S}_k] \mathbb{P}((X_i^{pub}, X_i^{priv}) \in \mathcal{S}_k). \tag{7}$$

Proposition 2. *Under sophisticated advice weighting, the optimal weights λ_L^{SAW} and λ_H^{SAW} satisfy $\lambda_H^{SAW} < \lambda^{NAW} < \lambda_L^{SAW}$. Furthermore, $OPT^{SAW} \leq OPT^{NAW}$, where OPT^{SAW} and OPT^{NAW} represent the optimal squared errors of SAW and NAW, respectively.*

The first part of Proposition 2 shows that the sophisticated advice weighter adheres more to the algorithm in the partition where the expected squared impact of the private features is low and adheres less where it is high. Moreover, the naïve advice weighter's optimal weight falls between the sophisticated advice weighter's optimal weights for the two sets. The second part of Proposition 2 states that the naïve advice weighter's failure to differentiate among instances where there is less versus more impact of the private features degrades its performance. Putting the two parts together, we conclude that the naïve advice weighter underweights the algorithm when the impact of the private features is small and overweights the algorithm when the impact of the private features is large.

3.4. Naïve Advice Weighting Is Suboptimal Because the Best Prediction Is Often Outside the Advice-Weighting Region

In addition to being suboptimal because of overly constant weights, naïve advice weighting—like *any* advice-weighting strategy—is suboptimal because the best prediction may not be in the *advice-weighting region*. We define the advice-weighting region as the interval between the algorithm's prediction and the human's initial prediction, which corresponds to $\lambda_i \in [0,1]$. When the best prediction benchmark, y_i^*, does not lie within the advice-weighting region, no choice of $\lambda_i \in [0,1]$ can achieve the best prediction benchmark.

In what follows, we explore how often the best prediction falls in the advice-weighting region.

Definition 2. Prediction $\hat{Z}_i$ is median unbiased if $\mathbb{P}[\hat{Z}_i > Y_i^*] = \mathbb{P}[\hat{Z}_i < Y_i^*]$.

Proposition 3. *Let $\hat{Y}_i^{init}$ and/or $\hat{Y}_i^{alg}$ be median unbiased, and assume $\mathbb{P}[\hat{Y}_i^{init} = Y_i^*] = \mathbb{P}[\hat{Y}_i^{alg} = Y_i^*] = 0$. Then, Y_i^* is within the advice-weighting region with probability 1/2: that is,*

$$\mathbb{P}(\min\{\hat{Y}_i^{alg}, \hat{Y}_i^{init}\} \leq Y_i^* \leq \max\{\hat{Y}_i^{alg}, \hat{Y}_i^{init}\}) = \frac{1}{2}.$$

The median-unbiased and zero probability of perfect predictions assumptions is required to obtain a result of precisely 1/2. Figure 1 illustrates the general intuition behind the proof. For the best prediction to be inside the advice-weighting region, $\hat{y}^{alg}$ and $\hat{y}^{init}$ must lie on opposite sides of y^*. If $\hat{Y}^{init}$ and $\hat{Y}^{alg}$ are equally likely to err high versus low (and assuming that the two errors are independent from Assumption 2), then it is equally likely for them to err on opposite sides as it is for them to err on the same side. Thus, Proposition 3 demonstrates that often it is impossible for humans to achieve the best prediction via any advice-weighting strategy. Importantly, the human could use their private information to identify when advice weighting is suboptimal.

3.5. Hypotheses and Preview of Experiments

Our first study, presented in Section 4, is designed to test the key intuition developed from Propositions 1 and 2. To do this, we use three conditions in which participants experience (i) only instances with a low impact of the private features (analogous to $\mathcal{S}_L$), (ii) only instances with a high impact of the private features (analogous to $\mathcal{S}_H$), and (iii) a mixture of instances that contain both low and high impact of the private features (analogous to $\mathcal{S}$). In practice, the

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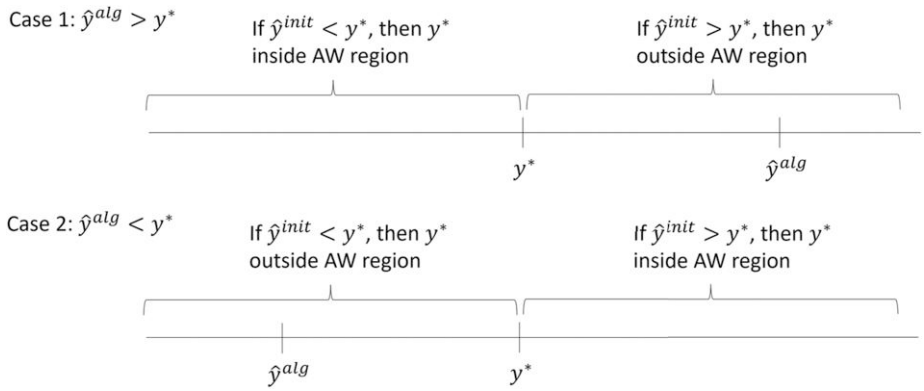
Figure 1. Depiction of When the Advice-Weighting (AW) Region Contains y^*



third condition—where humans occasionally have access to valuable private features—is most common, and we aim to empirically study whether participants in this condition are biased toward naïve advice weighting and ultimately, whether this leads to degradation in prediction accuracy. We give all participants access to historical instances consisting of (x_k^{pub}, x_k^{priv}), $\hat{y}_k^{alg}$, and y_k, and then, we elicit $\hat{y}_i^{init}$ and $\hat{y}_i^{final}$ for a series of new instances; this allows us to estimate participants' weight on the algorithm.

Hypothesis 1. *Relative to humans who experience instances of only high-impact private features or only low-impact private features, humans who experience a mix of both cases place a lower weight on the algorithm when the impact of the private features is low and place a higher weight on the algorithm when the impact of the private features is high.*

Hypothesis 2. *Relative to humans who experience instances of only high-impact private features or only low-impact private features, humans who experience a mix of both cases have worse final prediction accuracy for both cases.*

We hypothesize that one way to mitigate the overly constant weights problem of NAW would ideally be to tell decision makers which of the information available to them is public versus private. We study the impact of a practical intervention: providing *feature transparency*—telling humans which features the algorithm does take into account (i.e., which features are public). We believe that feature transparency will mitigate naïve advice-weighting behavior by helping humans recognize when they have impactful private features that warrant a substantial deviation from the algorithm, leading to the following two hypotheses that we test in a second study presented in Section 5.

Hypothesis 3. *Feature transparency increases humans' weight on the algorithm when the impact of the private features is low but decreases it when the impact of the private features is high.*

Hypothesis 4. *Feature transparency improves human's final prediction accuracy both when the impact of the private features is low and when it is high.*

Finally, Proposition 3 suggests that humans often need to deviate from the entire class of advice-weighting strategies altogether to achieve the best performance. We hypothesize that if—instead of advice weighting—humans anchor on the algorithm's prediction and make an adjustment to specifically account for only their private features, they may increase their likelihood of going outside the advice-weighting region in a way that improves performance; effectively, they can use their private features to predict when the advice-weighting region is suboptimal and correctly adjust in the other direction. We test this hypothesis in a third study presented in Section 6.

Hypothesis 5. *Nudging humans to follow a strategy in which they anchor on the algorithm and adjust based on their private features improves their final prediction accuracy.*

4. Study 1: Uncovering a Naïve Advice-Weighting Bias

Study 1 tests Hypotheses 1 and 2 in a controlled online experiment.[1]

4.1. Design

4.1.1. Participant Experience. Participants are tasked with predicting demand for new products. Each new product i is characterized by two features—"feature A" and "feature B"—where feature A corresponds to x_i^{pub} and feature B corresponds to x_i^{priv}. The outcome, Y_i, is the actual demand for product i. The following sequence of steps details the participant's experience; select screenshots are included in Online Appendix E.

1. *Instructions and comprehension checks.* Participants are introduced to the demand prediction task and objective of minimizing absolute prediction error, and they are tested for comprehension.

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2. *Historical data review.* Participants view historical data for 20 products, with the option to continue to view more historical data for as many products as they wish. For each product i, they observe x_i^{pub}, x_i^{priv}, and realized (actual) demand y_i.

3. *Demand predictions without algorithm.* Sequentially for each of $i = 1, \ldots, 20$ new products not seen in step 2, participants are given x_i^{pub} and x_i^{priv}, and they are asked for their demand prediction, $\hat{y}_i^{init}$. After predicting demand for product i, the participant is given the actual demand, y_i, and their absolute prediction error, $|\hat{y}_i^{init} - y_i|$.

4. *Algorithm introduction.* Participants are informed that an algorithm has been developed to help them predict demand; they are not given $f_{alg}(\cdot)$ or told that the algorithm only uses x_i^{pub}. To give participants experience with the algorithm, they are shown a summary table consisting of the following data for each of the 20 products from step 3: x_i^{pub}, x_i^{priv}, y_i, $\hat{y}_i^{alg}$, and both the algorithm's and their absolute prediction errors $|\hat{y}_i^{alg} - y_i|$ and $|\hat{y}_i^{init} - y_i|$.

5. *Demand predictions with algorithm.* Sequentially, for each of $i = 1, \ldots, 20$ new products not seen in step 2 or step 3, participants are first given only x_i^{pub} and x_i^{priv}, and they are asked for their demand prediction, $\hat{y}_i^{init}$. Then, the participant is given the algorithm's demand prediction, $\hat{y}_i^{alg}$, and asked for their final demand prediction, $\hat{y}_i^{final}$. Finally, they are told the actual demand, y_i, as well as both the algorithm's and their absolute prediction errors $|\hat{y}_i^{alg} - y_i|$ and $|\hat{y}_i^{init} - y_i|$.

Participants' demand predictions in step 3 ($\hat{y}_i^{init}$) and step 5 ($\hat{y}_i^{final}$) were incentivized for accuracy; participants received \$7 for completing the experiment (as in study 2 and study 3) plus an additional bonus of \$7 − \$0.15 × (root mean squared error). A majority of our analyses focuses on step 5, where participants have access to an algorithm to make their final demand predictions.

4.1.2. Behind the Scenes: Data Generation. For product i, actual demand is generated as

$$Y_i = 131 + 1.6X_i^{pub} + 0.75X_i^{priv} + \epsilon_i, \tag{8}$$

where ϵ_i is drawn from a normal distribution with mean of zero and standard deviation of four, X_i^{pub} is drawn from a discrete uniform distribution with support $\{20, 80\}$, and X_i^{priv} is drawn from zero-mean distributions that differ across our three conditions, which will be described in Section 4.1.3; for all three conditions, X_i^{pub} and X_i^{priv} are independent.

The algorithm's demand prediction is generated as

$$\hat{Y}_i^{alg} = 131 + 1.6X_i^{pub}. \tag{9}$$

From Definition 1, the impact of the private feature is $V_i \triangleq Y_i^* - \hat{Y}_i^{alg} = 0.75X_i^{priv}$.

Participants are not explicitly given Equations (8) and (9), nor are told how ϵ_i, X_i^{pub}, and X_i^{priv} are generated. That said, because participants have unlimited access to historical data in step 2, a participant could theoretically recover (8) and achieve the best prediction benchmark.[2]

4.1.3. Conditions. Participants are randomly assigned to one of the following three conditions, where the only difference across conditions is the distribution used to generate X_i^{priv} or equivalently, the impact of the private feature V_i. With slight abuse of language, we use "low" ("high") impact of the private feature to describe products with small (large) $|v_i|$.

1. *Always low impact of the private feature (Always Low $|v_i|$).* The values of X_i^{priv} are drawn from a discrete uniform distribution with support $\{-10, 10\}$. The impact of the private feature is low relative to the other conditions, with $|v_i| \in [0, 7.5]$; this leads to strong algorithm performance.

2. *Always high impact of the private feature (Always High $|v_i|$).* The values of X_i^{priv} are drawn from a discrete uniform distribution with support $\{-150, -50\} \cup \{50, 150\}$. The impact of the private feature is high relative to the other conditions, with $|v_i| \in [37.5, 112.5]$; this leads to poor algorithm performance.

3. *Mixed impact of the private feature (Mixed $|v_i|$).* Each value of X_i^{priv} is drawn from a discrete uniform distribution with support $\{-10, 10\}$ with probability 0.5 and from a discrete uniform distribution with support $\{-150, -50\} \cup \{50, 150\}$ with probability 0.5. In expectation, this leads to half of the products being identical to products in the first condition where the impact of the private feature is low and the other half of the products being identical to products in the second condition where the impact of the private feature is high.

It is helpful to cast our experimental design as a 2 × 2 mixed design (see Table 1). The first dimension is the impact of the private feature, which is either low or high. The second dimension is the participant's exposure set: whether the participant is exposed to a mixture of products with low and high impact of the private feature ("mixed" exposure set) or is exposed to only a single impact of the private feature—either always low or always high ("single" exposure set).

We are most interested in studying participants' predictions in the *Mixed* $|v_i|$ condition as this is reflective of practice where humans occasionally have access to valuable private features. We use participants in the other two conditions to represent benchmarks for sophisticated advice weighters—humans who make final predictions based on impact of the private features—because by construction, all of their predictions are based on a single exposure set. Note

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Table 1. How Our Three Conditions Achieve a 2×2 Mixed Design

	Exposure set	
Impact of the private feature	Mixed	Single
Low	*Mixed* $\lvert v_i \rvert$	*Always Low* $\lvert v_i \rvert$
High	*Mixed* $\lvert v_i \rvert$	*Always High* $\lvert v_i \rvert$

Note. Note that instances from the *Mixed* $\lvert v_i \rvert$ condition are divided between two cells depending on the instance's impact of the private feature, and these cells will be evaluated separately despite coming from participants in the same condition.

that these benchmarks are valid regardless of the sophistication of participants in these two conditions because for a single exposure set, naïve and sophisticated advice-weighting behaviors are approximately equivalent. By comparing across the columns in Table 1, we can identify whether humans experiencing a mixed exposure set can sufficiently distinguish instances with low and high impact of the private features (i.e., behave as sophisticated advice weighters) versus have a bias toward naïve advice weighting.

4.1.4. Dependent Variables. We use *median weight on algorithm (MedWOA)* as our dependent variable for Hypothesis 1. We first define participant j's *weight on algorithm* for product $i = 1, \ldots, 20$ in step 5 as

$$WOA_{ij} = \min\left(\max\left(\frac{\hat{y}_{ij}^{final} - \hat{y}_{ij}^{init}}{\hat{y}_i^{alg} - \hat{y}_{ij}^{init}}, 0\right), 1\right). \tag{10}$$

We note that WOA_{ij} is an estimate for λ_{ij} defined for advice-weighting behavior in (5). Following the advice-weighting literature, we note that WOA_{ij} is a winsorized value between zero and one, and we exclude WOA_{ij} if $\hat{y}_i^{alg} = \hat{y}_{ij}^{init}$. Subsequently, we define

$$MedWOA_j^L = \text{median}(WOA_{ij} \;\; \forall i \text{ s.t. } x_i^{priv} \in \{-10, 10\}); \tag{11}$$

$$MedWOA_j^H = \text{median}(WOA_{ij} \;\; \forall i \text{ s.t. } x_i^{priv} \in \{-150, -50\} \cup \{50, 150\}). \tag{12}$$

$MedWOA_j^L$ ($MedWOA_j^H$) can be interpreted as participant j's median value of WOA_{ij} for all products with low (high) impact of the private feature. We note that only participants in the *Mixed* $\lvert v_i \rvert$ condition will have values for both $MedWOA_j^L$ and $MedWOA_j^H$ because participants in the other conditions only experience products in a single exposure set.

We use *root median squared error (RMedSE)* as our dependent variable for Hypothesis 2. For each participant j and considering products $i = 1, \ldots, 20$ in step 5, we define

$$RMedSE_j^L = \sqrt{\text{median}([\hat{y}_{ij}^{final} - y_i]^2 \;\; \forall i \text{ s.t. } x_i^{priv} \in \{-10, 10\}}); \tag{13}$$

$$RMedSE_j^H = \sqrt{\text{median}([\hat{y}_{ij}^{final} - y_i]^2 \;\; \forall i \text{ s.t. } x_i^{priv} \in \{-150, -50\} \cup \{50, 150\}}). \tag{14}$$

4.2. Results

Our analyses include data from 359 participants from Mechanical Turk who passed two comprehension checks and completed the study.[3] By randomly assigning participants across conditions, we had 119 participants in the *Always Low* $\lvert v_i \rvert$ condition, 121 in the *Always High* $\lvert v_i \rvert$ condition, and 119 in the *Mixed* $\lvert v_i \rvert$ condition. The mean time to complete the study was 31.55 minutes, and the mean bonus payment was \$1.75. In what follows, we report our core results in Sections 4.2.1 and 4.2.2 and present supplementary analyses in Online Appendix B as outlined in Section 4.3.

4.2.1. Weight on Algorithm Results. Figure 2 summarizes the results on participants' median weight on algorithm. Recall that participants in the *Always Low* $\lvert v_i \rvert$ and *Always High* $\lvert v_i \rvert$ conditions represent benchmarks for sophisticated advice weighters—humans who make predictions based on the impact of the private features—because by construction, all of their predictions are for a single exposure set. As expected, these participants place more weight on the algorithm when they are only exposed to products with a low impact of the private feature compared with when they are only exposed to products with a high impact of the private feature because the algorithm performs considerably better for products with a low impact of the private feature ($t(237.92) = -12.72, p = 0.00$).

Our primary interest is studying behavior of participants in the *Mixed* $\lvert v_i \rvert$ condition. As detailed in Hypothesis 1, we hypothesize that participants in this condition would be unable to sufficiently distinguish between instances with low and high impact of the private features and would instead be biased toward naïve advice weighting. We perform two one-sided t tests comparing mean values of *MedWOA* across each row in Table 1. For convenience, let $\mathcal{C}_L$, $\mathcal{C}_H$, and $\mathcal{C}_M$ be the sets of participants assigned to the *Always Low* $\lvert v_i \rvert$, *Always High* $\lvert v_i \rvert$, and *Mixed* $\lvert v_i \rvert$ conditions, respectively.

For the first part of Hypothesis 1, we test whether

$$\frac{\sum_{j \in \mathcal{C}_M} MedWOA_j^L}{\lvert \mathcal{C}_M \rvert} \leq \frac{\sum_{j \in \mathcal{C}_L} MedWOA_j^L}{\lvert \mathcal{C}_L \rvert} \tag{15}$$

(i.e., considering only products with a low impact of the private feature, whether participants exposed to a mixed exposure set place less weight on the algorithm than participants exposed to a single exposure set). As shown in the left two bars in Figure 2, for low impact of the private feature products, participants in the *Mixed* $\lvert v_i \rvert$ condition had a significantly smaller mean

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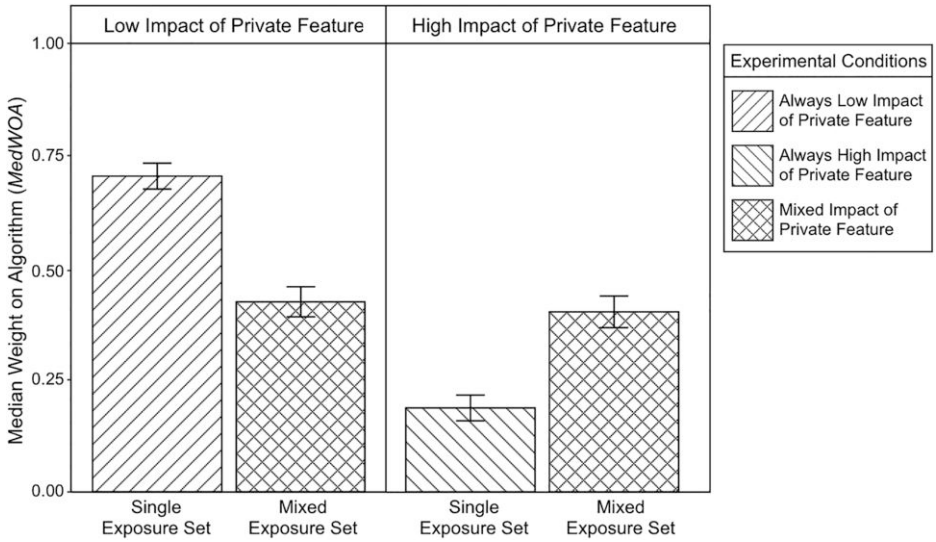
Figure 2. Median Weights on Algorithm Results Are Averaged (Mean) by Exposure Set Separately for Low and High Impact of the Private Feature



Note. Standard error bars are shown.

$MedWOA^L$ compared with participants in the *Always Low* $|v_i|$ condition ($t(230.83) = -6.33, p = 0.00$).

For the second part of Hypothesis 1, we test whether

$$\frac{\sum_{j\in\mathcal{C}_M} MedWOA_j^H}{|\mathcal{C}_M|} \geq \frac{\sum_{j\in\mathcal{C}_H} MedWOA_j^H}{|\mathcal{C}_H|} \tag{16}$$

(i.e., considering only products with a high impact of the private feature, whether participants exposed to a mixed exposure set place more weight on the algorithm than participants exposed to a single exposure set). As shown in the right two bars in Figure 2, for high impact of the private feature products, participants in the *Mixed* $|v_i|$ condition had a significantly larger mean $MedWOA^H$ compared with participants in the *Always High* $|v_i|$ condition ($t(227.53) = 4.72, p = 0.00$).

In addition to conducting t tests that study behavior on products with low and high impact of the private features separately, we can also test how the difference in average *MedWOA* between products with low and high impact of the private features compares across participants who experience a mixed versus single exposure set. A bias toward naïve advice weighting should result in a smaller difference in average *MedWOA* for participants experiencing a mixed exposure set: that is,

$$\frac{\sum_{j\in\mathcal{C}_M} MedWOA_j^L}{|\mathcal{C}_M|} - \frac{\sum_{j\in\mathcal{C}_M} MedWOA_j^H}{|\mathcal{C}_M|} \leq \frac{\sum_{j\in\mathcal{C}_L} MedWOA_j^L}{|\mathcal{C}_L|} - \frac{\sum_{j\in\mathcal{C}_H} MedWOA_j^H}{|\mathcal{C}_H|}. \tag{17}$$

When we regress the *MedWOA* for each participant on impact of the private feature interacted with exposure set, clustering standard errors by participant, we find a significant positive coefficient on the interaction term ($\beta = 0.49, p = 0.00$). In other words, the difference in average *MedWOA* between products with low versus high impact of the private feature is larger when participants experience a single exposure set than a mixed set; see Online Appendix B.1 for details. In fact, we find that for participants who experience a mixed exposure set, their average *MedWOA* is not significantly different for products with a low versus high impact of the private feature ($t(235.44) = -0.46, p = 0.34$).

Our results confirm Hypothesis 1 by showing that participants who experience a mixed exposure set underweight the algorithm when the impact of the private feature is low and overweight the algorithm when the impact of the private feature is high compared with participants who experience a single exposure set (representing hypothetical sophisticated advice weighters who fully differentiate between the sets). This provides empirical evidence supporting our proposed conjecture that humans are biased toward naïve advice weighting, insufficiently distinguishing when they should place more/less weight on an algorithm as a function of the impact of the private features.

4.2.2. Prediction Error Results. Figure 3 summarizes results showing the impact of the naïve advice-weighting bias on participants' root median squared error (*RMedSE*). As expected, participants in the *Always High* $|v_i|$ condition have larger prediction error compared with participants in the *Always Low* $|v_i|$ condition because the algorithm provides considerably less value

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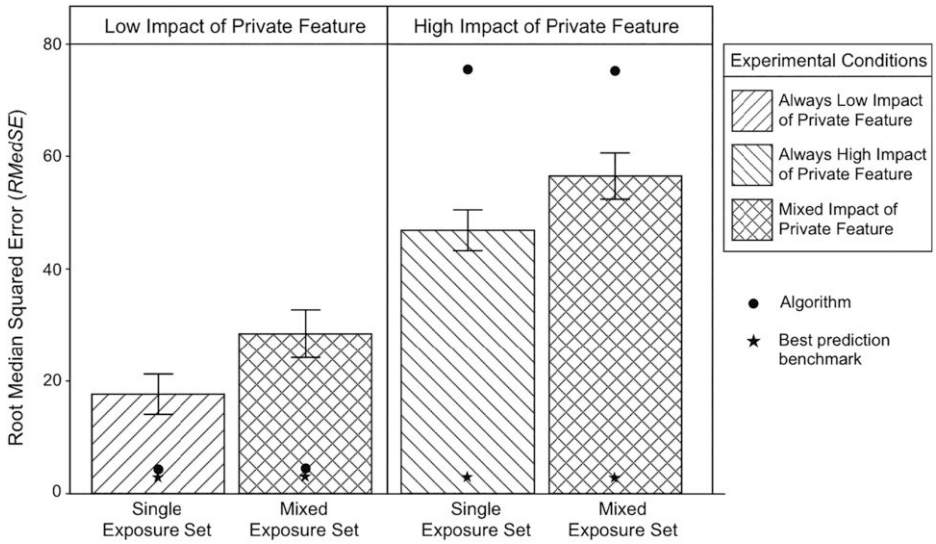
Figure 3. Root Median Squared Error Results Are Averaged (Mean) by Exposure Set Separately for Low and High Impact of the Private Feature



Notes. Standard error bars are shown. Additionally, the mean *RMedSE* is reported for the algorithm and the best prediction benchmark.

when the impact of the private feature is high ($t(238.00) = 5.74, p = 0.00$). Similarly, participants in the *Mixed* $|v_i|$ condition have larger prediction error on products with high impact of the private feature compared with their prediction error on products with low impact of the private feature ($t(235.66) = 4.73, p = 0.00$).

Our primary interest is studying the prediction error of participants in the *Mixed* $|v_i|$ condition to understand how their bias toward naïve advice weighting impacts their prediction error. As detailed in Hypothesis 2, we hypothesize that these participants will perform worse compared with sophisticated advice-weighter benchmarks for both subsets of products with low and high impact of the private feature. We perform two one-sided t tests comparing mean values of *RMedSE* across each row in Table 1. First considering only products with a low impact of the private feature, we test whether participants who experience a mixed exposure set have larger prediction error than participants who experience a single exposure set; specifically, we test whether

$$\frac{\sum_{j\in\mathcal{C}_M} RMedSE_j^L}{|\mathcal{C}_M|} \geq \frac{\sum_{j\in\mathcal{C}_L} RMedSE_j^L}{|\mathcal{C}_L|}. \tag{18}$$

As shown in the left two bars in Figure 3, participants in the *Mixed* $|v_i|$ condition had a significantly larger mean $RMedSE^L$ versus those in the *Always Low* $|v_i|$ condition ($t(229.08) = 1.94, p = 0.03$).

Next considering only products with a high impact of the private feature, we test whether participants who experience a mixed exposure set have larger prediction error than participants who experience a single exposure set; specifically, we test whether

$$\frac{\sum_{j\in\mathcal{C}_M} RMedSE_j^H}{|\mathcal{C}_M|} \geq \frac{\sum_{j\in\mathcal{C}_H} RMedSE_j^H}{|\mathcal{C}_H|}. \tag{19}$$

As shown on the right side of Figure 3, participants in the *Mixed* $|v_i|$ condition had a significantly larger $RMedSE^H$ versus those in the *Always High* $|v_i|$ condition ($t(233.67) = 1.76, p = 0.04$).

Together, our results confirm Hypothesis 2 by showing that participants who experience a mixed exposure set perform worse for both subsets of products with low and high impact of the private features compared with sophisticated advice-weighter benchmarks. Thus, our results provide empirical evidence that a bias toward naïve advice weighting negatively impacts prediction accuracy.

4.3. Additional Analyses and Discussion

We report several supplementary analyses in Online Appendix B. These include (1) replicating results at the task level, (2) observing minimal learning effects over time, (3) verifying robustness to unwinsorized *WOA*, (4) studying drivers of *WOA*, (5) conducting mediation analysis, (6) describing the performance of participants' demand predictions without the algorithm (in step 3) to show how it compares with the algorithm, (7) examining participants' initial predictions that precede seeing the algorithm (in step 5) to show that their accuracy is not significantly different from predictions without the algorithm (in step 3), (8) reporting task completion time statistics, and (9) examining how participants' variance in their observed *WOA* across all products

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compares to the variance in their optimal *WOA* if they were fully sophisticated advice weighters.

Study 1 provides evidence supporting a bias toward *naïve advice weighting* as opposed to sophisticated advice-weighting behavior; people take an overly constant weighted average of the algorithm's predictions with their initial predictions, not differentiating enough based on their private features. As outlined in the theory presented in Section 3.3, our empirical results confirm that this bias leads to predictable patterns of over- and underadherence to the algorithm and degrades performance when people face a mixed exposure set.

Although this study was designed to test the theory presented in Section 3.3, recall from Section 3.4 that the naïve advice-weighting strategy is suboptimal not only because the weight on advice is overly constant but also, because the best prediction benchmark (Y_i^*) may not be in the advice-weighting region (see, e.g., Figure 1). Similar to what Proposition 3 predicted, we find that in Study 1, the best prediction benchmark falls within participants' advice-weighting regions in 55% of instances. In contrast, participants' actual final predictions $\hat{y}_i^{final}$ fall within their advice-weighting regions in 84% of instances in the *Always Low* $|v_i|$ condition, 83% of instances in the *Always High* $|v_i|$, and 87% of instances in the *Mixed* $|v_i|$; these large differences illustrate that people are advice weighting much more than they should be regardless of condition. Importantly, people could use their private features to realize that sometimes, y_i^* is not in the advice-weighting region; in our following studies, we aim to focus people on their private features in hopes to move away from naïve advice-weighting behavior.

5. Study 2: The Impact of Transparency on Naïve Advice Weighting

What can system designers do to mitigate naïve advice-weighting behavior? Study 1 demonstrates that extensive algorithm performance feedback is not enough for people to figure out when their private information warrants a large or small deviation from the algorithm. How can system designers help people with bounded cognitive ability with this issue? Of course, by its very nature, system designers often do not know peoples' private information. However, as detailed in Hypotheses 3 and 4, providing *feature transparency* (training humans about which features the algorithm *does* take into account) may help mitigate naïve advice-weighting behavior by improving their ability to recognize when they have impactful private features, leading to improved performance accuracy for both products with low and high impact of the private features. Note that such a feature transparency intervention can be readily implemented across a wide variety of algorithms and industries because it only requires communicating the algorithm's input features and is independent of the complexity of the underlying algorithm. In practice, if system designers *were* aware of the private features, communicating this information would likely be even more helpful.

We contrast this insight-inspired *feature transparency* intervention with another unrelated *training data transparency* intervention, in which we describe in more detail how much data the algorithm uses in its training process. Similar types of transparency have been shown to increase overall trust in algorithms (e.g., see Anik and Bunt 2021 and Balayn et al. 2022). However, we hypothesize that it will not be effective in mitigating naïve advice weighting because it is not designed to help participants effectively discriminate between situations where they should versus should not adhere to the algorithm. We include this *training data transparency* manipulation as an additional control condition to disentangle the effect of providing people with knowledge of public information from the effect of providing people with more knowledge of the algorithm in general, which may also cause them to more carefully use the algorithm's forecasts.

5.1. Design

The participant experience, data generation, and dependent variables are identical to the *Mixed* $|v_i|$ condition in Study 1, except for the additions outlined in the three treatment conditions defined below.[4] Notably, these conditions only differ by information shared when introducing the algorithm; thus, both the algorithm and the best prediction benchmark each have identical performance across all conditions.

1. *No transparency.* This condition is identical to the *Mixed* $|v_i|$ condition in Study 1.

2. *Feature transparency.* We add the following language when introducing the algorithm in step 4: "The company has informed you that the algorithm uses only feature A to make its demand predictions". Recall that feature A corresponds to x_i^{pub} in our model.

3. *Training data transparency.* We add the following language when introducing the algorithm in step 4: "The company has informed you that the algorithm was trained on a data set of 9,834 products."

In both the *feature transparency* and *training data transparency* conditions, we add a comprehension check question verifying that participants understood the transparency description. We also remind participants of the transparency description when predicting demand with the algorithm for each of the 20 products in step 5; screenshots are included in Online Appendix F. For convenience, we define $\mathcal{C}_{NT}$, $\mathcal{C}_{FT}$, and $\mathcal{C}_{TDT}$ to be the set of participants assigned to the *no transparency*, *feature transparency*, and *training data transparency* conditions, respectively.

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5.2. Results

Our analyses include data from 521 Prolific participants who passed the comprehension check criteria by answering at least three of five questions correctly on their first try.[5] By randomly assigning participants across conditions, we had 172 participants in the *no transparency* condition, 171 in the *feature transparency* condition, and 178 in the *training data transparency* condition. The mean study completion time was 30.80 minutes, and the mean bonus payment was $1.54. Supplementary analyses can be found in Online Appendix C as outlined in Section 5.3.

5.2.1. Weight on Algorithm Results. Figure 4 summarizes the results on participants' median weight on algorithm (*MedWOA*). We are most interested in how the difference in average *MedWOA* between products with low and high impact of the private features compares across participants who are provided feature transparency versus no transparency. If feature transparency indeed mitigates the bias toward naïve advice weighting, we should see a larger difference in average *MedWOA* for participants provided feature transparency: that is,

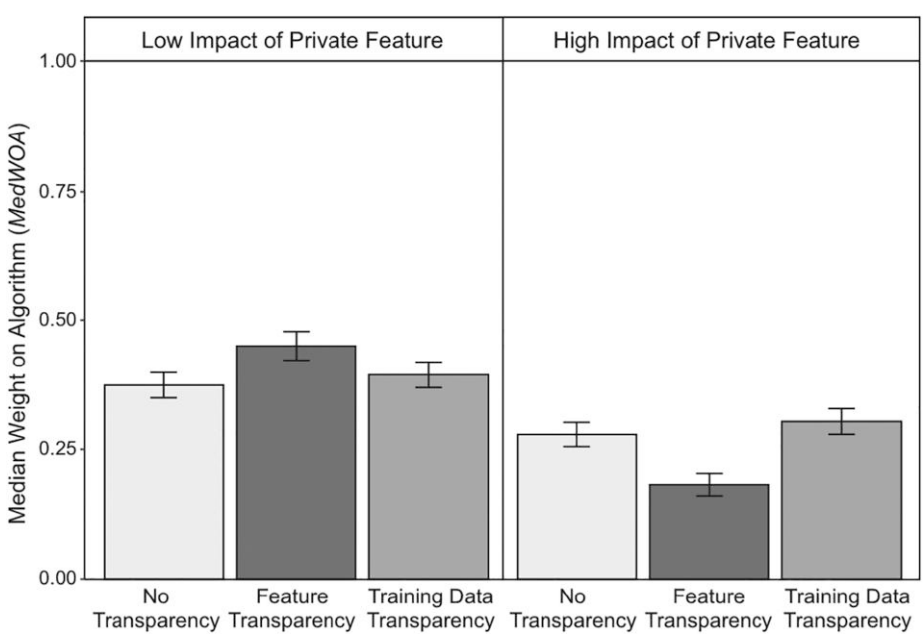
$$\frac{\sum_{j\in\mathcal{C}_{FT}} MedWOA_j^L}{|\mathcal{C}_{FT}|} - \frac{\sum_{j\in\mathcal{C}_{FT}} MedWOA_j^H}{|\mathcal{C}_{FT}|} \geq \frac{\sum_{j\in\mathcal{C}_{NT}} MedWOA_j^L}{|\mathcal{C}_{NT}|} - \frac{\sum_{j\in\mathcal{C}_{NT}} MedWOA_j^H}{|\mathcal{C}_{NT}|}. \quad (20)$$

When we regress the *MedWOA* for each participant on the impact of the private feature interacted with transparency type, clustering standard errors by participant, we find a significant coefficient on the interaction term ($\beta = 0.17, p = 0.00$), supporting Hypothesis 3. Similarly, we find that feature transparency mitigates naïve advice-weighting behavior more than training data transparency; namely, we repeat the same analysis as above, replacing no transparency with training data transparency ($\beta = 0.18, p = 0.00$). Finally (as an ex post test), we find that training data transparency does not significantly mitigate NAW behavior ($\beta = -0.01$, $p = 0.87$); see Table 2 for details.

Our results confirm Hypothesis 3 by showing that feature transparency mitigates naïve advice-weighting behavior by helping humans recognize when they have versus do not have impactful private features that warrant a substantial deviation from the algorithm. This is in contrast to many of the interventions in the transparency literature—like training data transparency—which do not aim to help people discriminate between situations where they should versus should not adhere to the algorithm; indeed, we find that training data transparency is not effective in mitigating NAW behavior.

5.2.2. Prediction Error Results. Figure 5 summarizes results showing the impact of transparency type on participants' root median squared error. As expected, within each condition, participants have larger prediction error

Figure 4. Median Weights on Algorithm Results Are Averaged (Mean) by Transparency Type Separately for Low and High Impact of the Private Feature



Note. Standard error bars are shown.

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Table 2. Regression Analyses of *MedWOA* by Impact of the Private Feature and Transparency Type

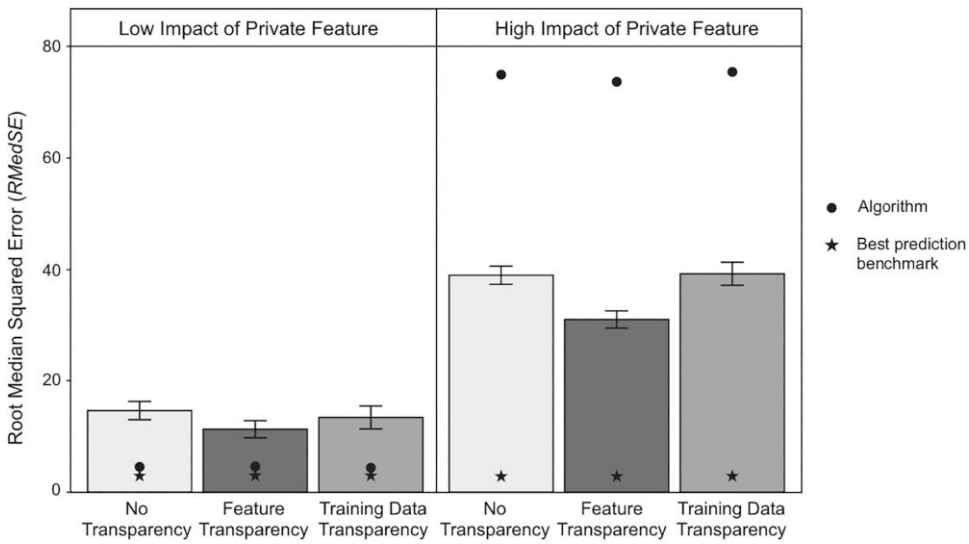
	Dependent variable: *MedWOA*		
	Model		
	Feature vs. no transparency	Feature vs. training data	Training data vs. no transparency
Variables			
(Intercept)	0.2786***	0.3038***	0.2786***
	(0.0234)	(0.0249)	(0.0234)
Low $\lvert v_i \rvert$	0.0953***	0.0902***	0.0953***
	(0.0205)	(0.0231)	(0.0205)
Feature Transparency	−0.0967***	−0.1219***	
	(0.0319)	(0.0330)	
Low $\lvert v_i \rvert \times$ *Feature Transparency*	0.1727***	0.1778***	
	(0.0352)	(0.0368)	
Training Data Transparency			0.0252
			(0.0342)
Low $\lvert v_i \rvert \times$ *Training Data Transparency*			−0.0050
			(0.0309)
Fit statistics			
Observations	686	698	700
R^2	0.09028	0.08604	0.02153
Adjusted R^2	0.08627	0.08209	0.01732

Note. Clustered (participant) standard errors are in parentheses.
***Significance code: 0.01.

on products with high impact of the private feature versus low impact of the private feature because the algorithm provides considerably less value when the impact of the private feature is high.

Our primary interest is studying how the prediction error of participants who are provided feature transparency compares with that of participants given no transparency. As detailed in Hypothesis 4, we hypothesize that feature transparency will lead to smaller prediction error for both subsets of products with low and high impact of the private feature. We first consider only products with a low impact of the private feature, and we use a one-sided t test to test whether participants provided with feature transparency have

Figure 5. Root Median Squared Error Results Are Averaged (Mean) by Transparency Type Separately for Low and High Impact of the Private Feature



Notes. Standard error bars are shown. Additionally, the mean *RMedSE* is reported for the algorithm and the best prediction benchmark.

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smaller prediction error than participants provided with no transparency:

$$\frac{\sum_{j\in\mathcal{C}_{FT}} RMedSE_j^L}{|\mathcal{C}_{FT}|} \leq \frac{\sum_{j\in\mathcal{C}_{NT}} RMedSE_j^L}{|\mathcal{C}_{NT}|}. \tag{21}$$

As shown in Figure 5, participants in the *feature transparency* condition had a significantly smaller mean *RMedSE* compared with participants in the *no transparency* condition for products with low impact of the private feature ($t(340.99) = 2.72, p = 0.00$).

We next repeat this analysis on products with a high impact of the private feature; we test whether

$$\frac{\sum_{j\in\mathcal{C}_{FT}} RMedSE_j^H}{|\mathcal{C}_{FT}|} \leq \frac{\sum_{j\in\mathcal{C}_{NT}} RMedSE_j^H}{|\mathcal{C}_{NT}|}. \tag{22}$$

As shown in Figure 5, participants in the *feature transparency* condition had a significantly smaller *RMedSE* compared with participants in the *no transparency* condition for products with high impact of the private feature ($t(339.98) = 3.54, p = 0.00$). When considering overall *RMedSE* across products with both low and high impact of the private feature, we also find that participants in the *feature transparency* condition had a significantly smaller *RMedSE* compared with participants in the *no transparency* condition ($t(334.38) = 4.11, p = 0.00$).

Similarly, we can compare prediction errors across participants who are provided feature transparency versus training data transparency. We find that participants in the *feature transparency* condition had a smaller mean *RMedSE* compared with participants in the *training data transparency* condition for products with low impact of the private feature ($t(303.85) = 1.33, p = 0.09$) as well as for products with high impact of the private feature ($t(342.58) = 3.19, p = 0.00$); considering overall *RMedSE*, we find that feature transparency leads to significant improvements ($t(283.66) = 2.55, p = 0.01$). As expected, training data transparency is less effective in mitigating naïve advice-weighting behavior because it is not designed to help participants effectively discriminate between situations where they should versus should not adhere to the algorithm.

Together, our results confirm Hypothesis 4 by showing that participants who are provided feature transparency perform better for both subsets of products with low impact and high impact of the private features compared with participants given no transparency or training data transparency.

5.3. Additional Analyses and Discussion

We report supplementary analyses in Online Appendix C, including showing robustness of our findings using an alternative dependent variable and no differences in completion times across conditions. In addition, we included an open-ended question at the end of the study asking participants to explain their decision-making process. Ex post text analysis indicates that *feature transparency* causes people to be 32% less likely to mention "averaging" but 31% more likely to mention "adjusting." Moreover, regression analysis reveals that people who mention "adjusting" had an 18% lower prediction error and suffered significantly less from NAW than participants who mention "averaging."

Study 2 shows that providing feature transparency helps to mitigate NAW behavior more effectively than other types of algorithm transparency (e.g., training data transparency) that may uniformly increase humans' adherence to the algorithm. That said, similar to Study 1, the best prediction benchmark fell outside the advice-weighting region 52% of the time, yet participants' final predictions fell within their advice-weighting regions in 92% of instances under *no transparency*, 93% of instances under *training data transparency*, and 86% of instances under *feature transparency*. This suggests that our *feature transparency* intervention only modestly helps humans move away from advice-weighting strategies; rather, the benefit of the intervention seems to primarily be driven by helping humans move away from applying overly constant weights.

6. Study 3: Improving Feature Transparency

Because of our findings in Study 2, we are motivated to improve upon feature transparency by helping humans move away from advice-weighting strategies. We hypothesize that—in addition to feature transparency—nudging humans to follow a strategy in which they anchor on the algorithm and adjust based only on their private features may further improve their use of algorithmic predictions. Although encouraging (nudging) employees to follow algorithmic advice is common in practice (Sanders and Wood 2020), our nudge is novel in that it specifically builds upon feature transparency to nudge humans to differentially use the algorithm across instances with different impacts of the private features. Just like our feature transparency intervention, the addition of the nudge can be readily implemented across a wide variety of algorithms and industries because it is agnostic to the underlying algorithm (other than its inputs) and does not require any specific knowledge of the private features. Study 3 tests this improved feature transparency intervention in an online experiment.

We additionally use Study 3 to generalize some of our prior results to an experimental setup with several key differences from Studies 1 and 2. In Study 3, we situate participants in a naturalistic setting where they make demand forecasts for clothing items using information that they have intuition around (e.g., price and advertising spend). We use a more realistic demand model that cannot easily be intuited, and we increase the complexity of the structure of the private feature (ad spend), generating it according to a nonnegative

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random variable with a nonzero mean. Finally, we do not elicit participants' initial predictions when making predictions with the algorithm.

6.1. Design

All participants encounter data generated according to a mixed impact of the private feature, with three treatment conditions that differ by information shared when introducing the algorithm.[6]

6.1.1. Conditions.

1. *No transparency.* This is similar to the *no transparency* condition in Study 2.

2. *Feature transparency.* This is similar to the *feature transparency* condition in Study 2. Participants are informed that the algorithm uses only price to make its demand predictions.

3. *Feature transparency with adjusting nudge (adjusting nudge).* Identical to the *feature transparency* condition, participants are additionally nudged toward using a strategy of anchoring on the algorithm's predictions and adjusting them using only private features.

6.1.2. Behind the Scenes: Data Generation.

For clothing item i, actual demand is generated as

$$Y_i = \begin{cases} 1{,}250(X_i^{pub}-25)^{-0.3}+70+1.2X_i^{priv}+\epsilon_i & X_i^{pub}\le 99 \\ 1{,}250(X_i^{pub}-70)^{-0.3}-45+1.2X_i^{priv}+\epsilon_i & 99<X_i^{pub}\le 149, \\ 1{,}250(X_i^{pub}-120)^{-0.3}-160+1.2X_i^{priv}+\epsilon_i & 149<X_i^{pub} \end{cases}$$

where ϵ_i is drawn from a normal distribution with mean of zero and standard deviation of five and where X_i^{pub} represents price and is randomly sampled from $\{54, 59, 64, 69, \ldots, 194, 199\}$, reflecting common prices for clothing items. Here, X_i^{priv} represents advertising spend and is drawn from a discrete uniform distribution with support {90, 110} with probability 0.5 and {0, 50} ∪ {150, 200} with probability 0.5. An example of this type of disjoint ad spend is an influencer marketing budget, with nanoinfluencers charging \$10–\$100, mid-tier influencers charging \$500–\$5,000, and megainfluencers charging \$10,000+ per sponsored post (Hitchcock 2024). The demand model mimics a constant elasticity demand function where demand exponentially decreases as price increases, with discontinuities leading to larger drops in demand as price increases from \$99 to \$100 and from \$149 to \$150.

The algorithm's demand prediction is generated by the equation

$$\hat{Y}_i^{alg} = \begin{cases} 1{,}250(X_i^{pub}-25)^{-0.3}+190 & X_i^{pub}\le 99 \\ 1{,}250(X_i^{pub}-70)^{-0.3}+75 & 99<X_i^{pub}\le 149. \\ 1{,}250(X_i^{pub}-120)^{-0.3}-40 & 149<X_i^{pub} \end{cases}$$

Because the algorithm does not observe X_i^{priv} in its historical training data, the algorithm essentially subsumes the average impact of X_i^{priv} on demand, $1.2 * \mathbb{E}[X_i^{priv}] = 1.2 \times 100$, in its constant term. In fact, the constant terms in $\hat{Y}_i^{alg}$ (+190, +75, and –40 depending on the value of X_i^{pub}) can be written as $+70 + 1.2 \times 100$, $-45 + 1.2 \times 100$, and $-160 + 1.2 \times 100$, respectively. Thus, the impact of the private feature is $V_i = 1.2(X_i^{priv} - 100)$. For our analyses, we will consider clothing items to have a "low impact of the private feature" if their X_i^{priv} is between \$90 and \$110, whereas clothing items with a "high impact of the private feature" are those with X_i^{priv} between \$0 and \$50 or between \$150 and \$200.

6.1.3. Participant Experience.

Study 3 follows a similar participant experience as in Study 2 but with the following key changes; screenshots are included in Online Appendix G.

In *step 4: algorithm introduction*, all participants are shown a row below the summary table with the average value for each displayed column (e.g., average price, average ad spend, etc.). Participants in the *feature transparency* and *feature transparency with adjusting nudge* conditions are told: "The company has informed you that the algorithm uses only price to make its demand forecasts. It doesn't have access to any other information." Participants in the *feature transparency with adjusting nudge* condition are also told the following.

> The algorithm optimally uses the price information, but you may have extra information beyond price that the algorithm doesn't have access to. The algorithm assumes an average value of any extra information when making a forecast; therefore the algorithm makes great forecasts when your extra information is close to its average value, but not when your extra information is far from its average value. Therefore, we recommend you collaborate with the algorithm using the following strategy: Follow the algorithm when your extra information is close to its average value. Only override the algorithm if your extra information is far from its average value. If you override, focus on using your extra information to adjust the algorithm up or down.

In *step 5: demand predictions with algorithm*, participants are not asked for an initial demand prediction $\hat{y}_i^{init}$. We made this choice in order to try to prevent people in the *feature transparency with adjusting nudge* condition from anchoring on their initial prediction, and we chose to make this consistent across all three conditions to ensure that any differences that we see across conditions are not because of the absence of eliciting $\hat{y}_i^{init}$. For participants in the *feature transparency with adjusting nudge* condition, before asking for $\hat{y}_i^{final}$, we ask: "Do you have any extra information that is far from its average value?" If they select "No, I'll use the

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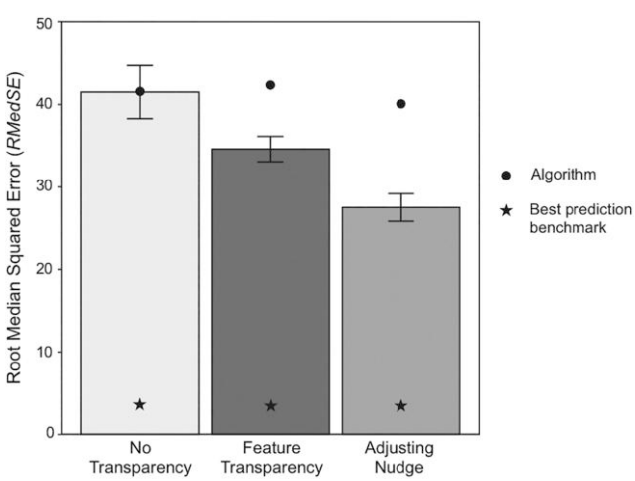
algorithm," then the algorithm's demand prediction, $\hat{y}_i^{alg}$, is recorded as their final demand prediction, $\hat{y}_i^{final}$. If they select "Yes, my extra information is far from its average value and I'd like to override the algorithm," then they are asked for their final demand prediction, $\hat{y}_i^{final}$. Here, participants are prompted with a strategy: "Use only your extra information to adjust the algorithm's forecast up or down."

6.2. Results

Our analyses include data from 549 Prolific participants who successfully passed the comprehension check criteria by answering at least three of four questions correctly on their first try.[7] By randomly assigning participants across conditions, we had 183 participants in each condition. The mean study completion time was 29.39 minutes, and the mean bonus payment was $1.37.[8]

We first present results showing the impact of transparency type on participants' root median squared error (Figure 6). Our primary interest is studying how the prediction error of participants who are provided with feature transparency paired with an adjusting nudge compares with that of participants given either no transparency or feature transparency alone. Using a one-sided t test, we find that participants in the *feature transparency with adjusting nudge* condition had a significantly smaller *RMedSE* compared with participants in the *no transparency* condition ($t(273.60) = 3.85, p = 0.00$). Additionally, using a one-sided t test, we find that participants in the *feature transparency with adjusting nudge* condition had a significantly smaller *RMedSE* compared with participants in the *feature transparency* condition ($t(361.44) = 3.11, p = 0.00$). Together, these results confirm Hypothesis 5, showing that participants who are provided with the adjusting nudge have lower prediction error compared with participants given no transparency or feature transparency alone. Furthermore, we replicate our results in Section 5.2.2 across a contextual setting and different data-generation process; participants who are provided with feature transparency have a lower prediction error compared with participants given no transparency ($t(260.90) = 1.94$, $p = 0.03$).

Figure 6. Root Median Squared Error Results Are Averaged (Mean) by Transparency Type



Notes. Standard error bars are shown. Additionally, the mean *RMedSE* is reported for the algorithm and the best prediction benchmark.

To understand how the adjusting nudge drives further improvement in decreasing participants' prediction error relative to feature transparency alone, we first examine whether participants provided with the adjusting nudge have a lower prediction error for clothing items with low impact of the private feature. Because our nudge explains that adjustments should only be made when the private feature is far from its average value, we would expect smaller adjustments—and therefore, better predictions—when the impact of the private features is low. Using a one-sided t test, we indeed find that participants in the *feature transparency with adjusting nudge* condition had a significantly smaller *RMedSE* on low impact of the private feature clothing items (*RMedSE* = 9.4) compared with participants in the *feature transparency* condition (*RMedSE* = 19.3) ($t(293.25) = 6.78, p = 0.00$). The graph of results split by low and high impact of the private features is included in Online Appendix D.1.

We next examine whether under the adjusting nudge, participants more frequently adjust the algorithm's predictions in the correct direction, indicating they are no longer advice weighting and instead, using only private features to adjust the algorithm. We define a participant's adjustment of the algorithm's prediction as being in the *correct direction* if and only if the following holds:

$$(\hat{y}_i^{alg} \geq \hat{y}_i^{final} \wedge \hat{y}_i^{alg} \geq y_i^*) \vee (\hat{y}_i^{alg} \leq \hat{y}_i^{final} \wedge \hat{y}_i^{alg} \leq y_i^*).$$

Participants provided with *no transparency* and *feature transparency* have average rates of making adjustments to the algorithm in the correct direction of 71.5% and 74.4%, respectively. Similar to what we observed in Study 2, this suggests that feature transparency only modestly helps humans move away from advice-weighting heuristics. In contrast, participants provided with the *feature transparency with adjusting nudge* have an average rate of adjustments in the correct direction of 93.5%, suggesting that the additional nudge successfully gets people to move away from advice weighting. A one-sided t test reveals that this improvement over *feature transparency* alone is significant ($t(302.53) = -14.30$, $p = 0.00$). See Online Appendix D.2 for more details.

Study 3 confirms our hypothesis that feature transparency can be improved by encouraging humans to use a strategy of anchoring on algorithmic predictions

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and adjusting based only on their private features. Our results suggest that our intervention—pairing feature transparency with this adjusting nudge—helps humans overcome *both* reasons why naïve advice weighting is suboptimal. Namely, it helps them better differentiate when they should adhere more or less to the algorithm, and it helps them adjust in the right direction more often than advice-weighting heuristics in general.

7. Conclusion

This paper proposes and provides experimental evidence that people's algorithm overrides are biased toward naïve advice weighting, taking a constant weighted average of the algorithm's prediction and their own initial prediction without accounting for the impact of the private features. This causes people to overadhere to the algorithm when they have highly valuable private information and underadhere to the algorithm when they do not as well as frequently adjust the algorithm in the incorrect direction. However, providing people with feature transparency and pairing it with education on adjusting algorithmic predictions based only on private information can help mitigate their bias toward NAW and improve predictive performance.

Our results generate insights for managers seeking to design algorithms and how they interface with humans. First, we help identify *when* human-algorithm collaborative performance is most hurt by NAW: when humans *sometimes* have valuable private information. In these settings, interventions that uniformly increase people's trust in the algorithm do not help address the underlying issue. Instead, interventions such as feature transparency—which help people discern when they have more or less valuable private information—are more appropriate. Because there are a plethora of types of algorithm transparency each with their own goals in changing how people interact with the algorithm, it is important for system designers to understand when they are in a situation which warrants addressing NAW rather than a different fundamental issue (e.g., incentive or trust issues).

Second, by illuminating *why* feature transparency and education on adjusting strategies help, our results provide some guiding objectives for algorithm developers, interface designers, and user trainers. Specifically, these activities should clearly communicate what features the algorithm uses, help people recognize when they have private information that warrants deviation, and facilitate adjustments based on such private information. These guidelines also correspond to natural limitations for feature transparency and our adjusting nudge; one should not expect them to work if—even with them—people cannot reasonably determine whether they have private information that warrants deviations or cannot make adjustments that improve performance even when they do know they have such private information. For example, a person encountering features that are highly correlated with public features may be tricked into *thinking* that they have important private information even when they do not. Or, a person may have access to a theoretically valuable private feature, but its relationship to the target is so convoluted that they are unable to translate it into adjustments that improve performance.

Finally, our results shed light on when system designers should let humans override algorithms in general. Giving humans this authority subjects the system to human cognitive limitations and noise, which leads some experts to suggest avoiding this setup (e.g., Kahneman et al. 2021). However, even when the organizational setting does not require a human to have final decision authority, our results suggest that we should still consider letting humans have override authority when they may have access to valuable private information. In such settings, letting humans override the algorithm can potentially add more value through incorporating their private information than harm by exposing the system to their noise and other biases. We also recommend that system designers work to identify, collect, and codify private information if possible. In general, we see research opportunity in improving our understanding of how to develop systems in which humans and algorithms focus and hone their relative strengths to enhance their long-run collaborative performance.

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Endnotes

[1] We preregistered our sample size, treatment conditions, data exclusion criteria, and planned analyses (see https://aspredicted.org/CN9_KTK). All statistical tests reported in the results were preregistered unless otherwise indicated.

[2] The data-generating processes were chosen so that demand was positive, varied enough to observe sufficient variation in participants' forecasts, and low enough so that we could cap participants' maximum forecast. In the mixed impact condition, low- and high-impact private feature products had an equal probability of appearing to maximize sample sizes for separate analyses. Low (high) impact of private feature products was calibrated such that the algorithm had a relatively high (low) accuracy versus human forecasts.

[3] In total, 534 Mechanical Turk workers attempted the study, each with a 99%+ approval rating and 1,000+ approvals. Three hundred and sixty participants then passed the comprehension checks, of which only one did not complete the study. Among the 359 participants, 208 were male, 256 had a bachelor's or advanced degree, 315 were White, and 185 had a yearly household income of $50,000 or more.

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[4] We preregistered our sample size, treatment conditions, data exclusion criteria, and planned analyses (see https://aspredicted.org/6KL_L8F). All statistical tests reported in the results were preregistered unless otherwise indicated.

[5] In total, 525 workers were recruited to complete the study, each with a 99%+ approval rating, 25+ previous submissions, and English listed as a fluent language. Among the 521 participants, 229 were male, 307 had a bachelor's or advanced degree, 412 were White, and 291 had a yearly household income of $50,000 or more.

[6] We preregistered our sample size, treatment conditions, data exclusion criteria, and planned analyses (see https://aspredicted.org/P7T_1WC). All statistical tests reported in the results were preregistered unless otherwise indicated.

[7] In total, 600 workers were recruited to complete the study who had not previously completed Study 2, each with a 99%+ approval rating, 25+ previous submissions, and English listed as a fluent language. Among the 549 participants, 275 were male, 321 had a bachelor's or advanced degree, 382 were White, and 342 had a yearly household income of $50,000 or more.

[8] Participants received a bonus of $7 − $0.08 × (root mean squared error) based on their demand predictions in steps 3 and 5.

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